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1. Overall Summary

This paper introduces DeLAMA, a novel decentralized and lifelong-adaptive multi-agent collaborative learning framework designed to overcome the limitations of centralized coordination and static collaboration relationships in multi-agent systems. DeLAMA empowers agents to autonomously identify beneficial collaborators and adapt to dynamic tasks by integrating a decentralized graph structure learning algorithm with a lifelong learning memory unit. The framework leverages algorithm unrolling to combine mathematical optimization with neural networks, enhancing expressive capabilities and computational efficiency. Theoretical analysis confirms communication efficiency and convergence rates. Experimental results on regression, image classification, and multi-robot mapping tasks demonstrate DeLAMA's superior performance, achieving a 98.80% reduction in MSE and a 188.87% improvement in classification accuracy compared to baselines. The system effectively discovers collaboration strategies and adapts to dynamic scenarios, making it a foundational technique for intelligent, decentralized, and dynamic multi-agent systems.

2. Main Idea

The central concept is DeLAMA, a decentralized and lifelong-adaptive multi-agent collaborative learning algorithm. It enables multiple agents to autonomously identify beneficial collaboration relationships and adapt to dynamically changing tasks over time, without relying on a central server. This is achieved by combining mathematical optimization and neural networks through algorithm unrolling.

3. Problem Being Solved

This paper addresses two main weaknesses in previous multi-agent collaborative learning systems: 1) the reliance on a central server for coordination, which introduces vulnerability (single point of failure) and high communication overhead; and 2) the use of predefined, static collaboration relationships, which are inflexible and inadequate for dynamic scenarios where tasks and agent relationships evolve over time.

4. Assumptions

- Local decision-making is done via training an individual model.
- Similar model parameters promote agents to collaborate.
- Training data arrives independently among agents at each time t .
- Data batches satisfy conditional independency on both agent-level and time-level.
- The adjacency matrix $W(t)$ is symmetric for Jacobi iteration.
- The linear model $f_{\theta}(t, \cdot)$ is linear and the loss function is a standard supervised training loss (for Theorem 1).
- The Hessian $H(\alpha_i(t))$ is Lipschitz continuous (for classification tasks).

5. Limitations

- Taylor expansion used in local learning can have potential approximation errors for non-linear functions, limiting the method's expressive capability for non-linear learning tasks.
- The original optimization problem has many hyperparameters, requiring significant effort for searching and tuning.
- The iterative algorithm requires numerous iterations, leading to significant communication overhead.
- In human-involved experiments, for harder collaboration problems with lower data quality, humans struggle to distinguish useful collaborators, suggesting DeLAMA's current collaboration model might not capture complex human logic and higher-order interactions.

6. Results and Conclusion

DeLAMA demonstrates superior performance across various tasks. In regression, it significantly outperforms federated learning and decentralized optimization methods, achieving a 98.80% reduction in MSE at the last timestamp ($t=10$). For image classification, DeLAMA effectively remembers previously learned data, outperforming classic lifelong learning approaches by 99.66% on MNIST and 134.01% on CIFAR-10 at $t=10$. It also improves classification accuracy by 54.42% on MNIST and 188.87% on CIFAR-10 compared to static collaboration graph methods. In multi-robot mapping, DeLAMA outperforms previous decentralized optimization methods, achieving a 95.31% F1-score at $t=12$. Theoretical analysis confirms that inter-agent collaboration is communication efficient and that the solution exhibits favorable convex optimization properties with quadratic and linear convergence rates. Human-involved experiments further validate that DeLAMA's collaboration mechanism mirrors human cognitive collaboration, showing a negative correlation between model parameter distance and human collaboration strength.

7. Real World Impact

Practical real-world applications of this research include vehicle-to-everything (V2X) communication-aided autonomous driving, multi-robot environment exploration, collaborative training of machine learning models across multiple clients (relevant for edge AI and smart healthcare), traffic control, and unmanned aerial vehicle (UAV) system control. It serves as a foundational technique for developing intelligent, decentralized, and dynamic multi-agent systems.

8. Graphs and Figures Analysis

Figure 1: The illustration of our decentralized and lifelong-adaptive multi-agent collaborative learning system.

This figure visually represents the DeLAMA system. It shows six agents, each with model parameters, observations, and collaboration relationships. The key aspect is that these elements (observations and collaboration relationships) are dynamic and adapt over time (Timestamp 1, 2, 3). The figure highlights that agents autonomously identify beneficial collaboration relationships and adapt to changing tasks without a central server.

Figure 2: Relationships among our lifelong collaborative learning and its related works: lifelong learning, federated learning, and multi-task relation learning.

This figure is a 2x2 matrix comparing DeLAMA with related works based on two dimensions: 'Dynamic scenario' (vs. 'Static scenario') and 'Predefined collaboration relations' (vs. 'Learned collaboration relations'). The top-left shows Lifelong learning (dynamic scenario, predefined relations), top-right shows DeLAMA (dynamic scenario, learned relations), bottom-left shows Federated learning (static scenario, predefined relations), and bottom-right shows Multi-task relation learning (static scenario, learned relations). It visually positions DeLAMA as unique in handling both dynamic scenarios and learning collaboration relationships.

Figure 3: The decentralized and lifelong-adaptive multi-agent collaborative learning system.

This figure illustrates the three main steps of the DeLAMA system: 1) Local learning, where each agent learns model parameters based on its own observations for initialization. 2) Collaboration relational inference, where agents transmit model parameters along the communication graph and learn collaboration relations. 3) Lifelong model update, where agents share model parameters with collaborators based on learned relations and refine their models. The figure shows that steps 2 and 3 are iterated until convergence, and highlights the communication graph constraint and collaboration relation.

Figure 4: The unrolled network structure of the collaboration system with input data at one exact time.

This figure depicts how the iterative optimization steps are unrolled into a deep neural network structure. Input data ($D_i(t)$) is fed into a local learning component (Φ_{local}) to initialize model parameters ($\theta_i(t,0)$). These initialized parameters, along with the communication graph ($C(t)$), are used by Φ_{graph} to infer collaboration relationships ($W(t)$). The inferred $W(t)$ and $\theta_i(t,0)$ are then used by Φ_{param} to update model parameters ($\theta_i(t,k)$). This process is iterated, showing the flow of information and how different components (local learning, graph inference, model update) are integrated into a single learnable network with learnable hyperparameters (γ).

Figure 5: Visualization of the learned graph structure at time $t = 3$ compared with the oracle collaboration structure.

This figure presents heatmaps visualizing the collaboration graph structures learned by different methods (GL-LogDet, GLasso, MTRL, L2G-PDS, NS, DeLAMA) at time $t=3$, alongside the Oracle structure. The intensity of the squares indicates collaboration strength between agents. DeLAMA's heatmap shows a structure very similar to the Oracle, indicating its ability to learn smooth and accurate edge weights, outperforming other baseline methods which show sparser or less accurate structures.

Figure 6: A collaboration result of two groups of agents that are performing two different regression tasks.

This figure consists of two parts. (a) Functions learned by each agent: Shows the regression lines learned by individual agents over time ($T=1$ to $T=5$). Initially, agents have limited knowledge, but as time progresses, their learned functions (colored lines) converge towards the true underlying functions (black lines), demonstrating lifelong adaptation. (b) Time evolving collaboration structure: Visualizes the collaboration graph among agents from $T=1$ to $T=5$. It shows that agents dynamically form two distinct collaboration groups, corresponding to the two different regression tasks they are performing. The connections strengthen and refine over time, indicating that DeLAMA enables the system to evolve with dynamic collaboration relationships.

Figure 7: Example of the evolving collaboration graphs.

This figure shows the time-evolving collaboration graphs for MNIST (a) and CIFAR-10 (b) datasets at different timestamps ($T=1, T=3, T=5, T=7$), compared to the Oracle (ground truth) structure. The heatmaps illustrate how DeLAMA dynamically adjusts collaboration relationships. Initially, agents might collaborate more broadly, but as observations grow, less useful relationships are pruned, and agents correctly identify and strengthen connections with useful collaborators, leading to a refined and accurate collaboration structure over time.

Figure 8: The simulation result of robot mapping at $t = 12$ based on the CubiCasa5k room Dataset.

This figure presents the robot mapping results at time $t=12$ for different decentralized collaborative model training approaches (DeLAMA, DeLAMA-WC, DSGD, DSGT, DiNNO) compared to the 'Truth Map'. Black areas represent empty space, and white areas represent occupied space. DeLAMA's map most closely resembles the 'Truth Map,' indicating superior performance in accurately mapping the environment through multi-agent collaboration, especially compared to methods that don't use collaboration (DeLAMA-WC) or other decentralized approaches.

Figure 9: The human evaluation of collaboration relationships with each other.

This figure shows three scatter plots, each representing a different data quality (High, Medium, Low). The x-axis represents human rating scores for collaboration (0=lowest, 5=highest), and the y-axis represents the logarithm of model parameter distance. The plots show a negative correlation: as human collaboration scores increase (meaning stronger collaboration), the model parameter distance between agents decreases. This supports the assumption that humans tend to collaborate with similar knowledge. For lower data quality, the spread of points is wider, indicating that humans find it harder to distinguish useful collaborators.

Figure 10: A time-evolving learning task example of two agents sampled from the regression task dataset.

This figure shows the learning progress of two agents over time ($T=1$ to $T=5$) on a regression task. Each subplot displays scatter points (training samples) and the agent's learned regression line (colored) compared to the ground truth function (black). Initially, agents have limited samples and their learned lines deviate from the ground truth. As time progresses and more samples are encountered, the agents' knowledge of the target function improves, and their learned lines converge closer to the ground truth, demonstrating the lifelong learning capability.

Figure 11: The human-machine interaction framework used for the human-involved experiment.

This flowchart illustrates the architecture for the human-involved experiment. It shows a 'Human machine interaction' layer (visual interaction, imitation) connecting 'Human' and 'Virtual agent' components. Below this, 'Background algorithms' (sharing model parameters) connect 'Proxy machine' and 'Machine' components. The 'Proxy machine' imitates human behaviors to enable interaction between humans and the DeLAMA-powered 'Machine' agents. This framework allows for real human evaluation of the collaboration algorithm.

Figure 12: The login page of the human-involved experiment, contains a user guide for the participants to learn how to play the game.

This figure displays a screenshot of the login page for the human-involved experiment. It includes fields for User ID and Username, and a 'How to play the game?' section that provides instructions to participants on the linear regression task, emphasizing that they only access a portion of data and can refer to other players' results to refine their answers.

Figure 13: The local learning page of the human-involved experiment.

This figure shows a screenshot of the local learning page in the human-involved experiment. It displays a scatter plot of data points and allows the participant to draw a line (representing their guess of the regression line). The equation of the drawn line ($y = 0.39x + -4.60$) and its parameters (K, b) are shown, which are based purely on the participant's visual information.

Figure 14: The aggregation page of the human-involved experiment, which contains other participants' results and the score rated by each agent.

This figure shows a screenshot of the aggregation page. It displays the current participant's learned line (red) along with the learning lines of other participants (blue, green). It also shows an evaluation score (0.1057) for the participant's performance, and a star rating system (e.g., User 2 has 5 stars), which allows agents to rate each other's collaboration. This page enables participants to see others' results and decide whether to modify their own.

Figure 15: The final page of the human-involved experiment with a leaderboard, each participant can look back to historical learning results.

This figure shows the final page of the human-involved experiment. It includes a 'Leaderboard' displaying average performance scores (e.g., 0.0397) and a 'Result Log' chart that plots the evaluation loss over time ($T=1$ to $T=4$) for different agents and the ground truth. This allows participants to review their historical learning results and compare them against the ground truth.